MATH 152 **QUIZ 9A** NAME Solutions

Mrs. Bonny Tighe 25 points

11.9 TA_____ Wed 11/14/2022

1. Find the power series representation for the following functions.

a) $f(x) = \dfrac{x}{1+3x^3}$ Find the interval of convergence for this series only.

5pts

$$\frac{x}{1-(-3x^3)} = \sum_{n=0}^{\infty} x(-3x^3)^n = \boxed{\sum_{n=0}^{\infty} (-1)^n 3^n x^{3n+1}}$$

$$|r| = |-3x^3| < 1 \qquad |3x^3| < 1 \qquad |x^3| < 1/3$$

$$-1/3 < x^3 < 1/3$$

$$\sqrt[3]{-1/3} < x < \sqrt[3]{1/3}$$

$$R = \sqrt[3]{1/3}$$

$$I = \left(-\sqrt[3]{1/3}, \sqrt[3]{1/3}\right)$$

Don't need to test end points because it is a geometric series

b) $f(x) = \ln(x+4)$

3pts

$$f'(x) = \frac{1}{x+4} = \frac{1/4}{1-(-x/4)} = \sum_{n=0}^{\infty} \frac{1}{4}\left(-\frac{x}{4}\right)^n =$$

$$f'(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{4^{n+1}} \qquad \int f'(x)\,dx = f(x)$$

$$\int \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{4^{n+1}}\,dx = C + \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{4^{n+1}(n+1)}$$

$$(C = \ln 4)$$

3 pts c) $f(x) = \dfrac{1}{(x-1)^2}$

$\int f(x)\,dx = \int (x-1)^{-2}\,dx = \dfrac{-1}{x-1}$

$\dfrac{-1}{x-1} = \dfrac{1}{1-x} = \sum_{n=0}^{\infty} x^n$

$\dfrac{d}{dx}\int f(x)\,dx = f(x)$

$\dfrac{d}{dx}\sum_{n=0}^{\infty} x^n = \boxed{\sum_{n=1}^{\infty} n x^{n-1}}$

2. Find the integral of the power series $\int \sum_{n=0}^{\infty} \dfrac{(-1)^{n+1}(x-2)^{n+2}}{5^n\,(2n)}\,dx$

1 pt $C + \sum_{n=0}^{\infty} \dfrac{(-1)^{n+1}(x-2)^{n+3}}{5^n(2n)(n+3)}$

3. Find the derivative of the given power series

$$f(x) = \sum_{k=0}^{\infty} \frac{(-1)^k (1\cdot 3\cdot 5\cdots(2k+1))\ x^{2k+1}}{3^{k+1}\,k!}$$

1 pt $f'(x) = \sum_{k=0}^{\infty} \dfrac{(-1)^k (1.3.5\cdots(2k+1))(2k+1)\,x^{2k}}{3^{k+1}\,k!}$

$$f(x) = \sum_{k=0}^{\infty} \frac{(-1)^k (1\cdot 3\cdot 5\cdots(2k+1))\ x^{2k}}{7^k\,k!}$$

1 pt $f'(x) = \sum_{k=1}^{\infty} \dfrac{(-1)^k (1.3.5\cdots(2k+1))\,2k\,x^{2k-1}}{7^k\,k!}$

5pts

4. Express the function as the sum of power series by first using partial fractions. Find the interval and radius of convergence. $f(x) = \dfrac{1+x}{2-5x+2x^2}$

$(2-x)(1-2x)$

$$\frac{A}{2-x} + \frac{B}{1-2x} = \frac{1+x}{(2-x)(1-2x)}$$

$$A(1-2x) + B(2-x) = 1+x$$

$x=2$ $\quad A(1-4) = 3 \quad A=-1$

$x = 1/2$ $\quad B(2-1/2) = 1+1/2$

$B(3/2) = 3/2$

$B = 1$

$$\frac{1}{1-2x} - \frac{1}{2-x}$$

$$\frac{1}{1-2x} - \frac{1/2}{1-x/2}$$

$$\sum_{n=0}^{\infty} (2x)^n - \frac{1}{2}(x/2)^n$$

$$\sum_{n=0}^{\infty}\left[2^n x^n - \frac{x^n}{2^{n+1}}\right] = \sum_{n=0}^{\infty} x^n\left[2^n - \frac{1}{2^{n+1}}\right]$$

$|r| = |2x|$ for one

$|2x| < 1$

$|x| < 1/2$

$-1/2 < x < 1/2$

$|r| = \left|\frac{x}{2}\right|$ for other

$\left|\frac{x}{2}\right| < 1$

$|x| < 2$

$-2 < x < 2$

they cross for interval

$I = (-1/2, 1/2)$

$R = 1/2$

5. Rewrite the series $f(x) = \sum_{n=1}^{\infty} \frac{(-1)^n \; x^{2n+1}}{3^{n+1} \; n!}$ here

2pt

$$\sum_{n=0}^{\infty} \frac{(-1)^{n+1} \; x^{2n+3}}{3^{n+2} \; (n+1)!}$$

6. Approximate the integral using power series and 4 non-zero terms.

$$\int_0^{0.2} \frac{x}{4+x^2}\,dx = \int \frac{\frac{x}{4}}{1+\frac{x^2}{4}}\,dx = \int_0^{0.2} \sum_{n=0}^{\infty} \frac{x}{4}\left(-\frac{x^2}{4}\right)^n dx$$

4 pts

$$\int_0^{0.2} \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{4^{n+1}}\,dx = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+2}}{4^{n+1}(2n+2)} \Bigg]_0^{0.2}$$

$$\frac{x^2}{4(2)} - \frac{x^4}{4^2(4)} + \frac{x^6}{4^3(6)} - \frac{x^8}{4^4(8)} \Bigg]_0^{0.2}$$

$n=0$ $\quad n=1$ $\quad n=2$ $\quad n=3$

$$\frac{(0.2)^2}{4(2)} - \frac{(0.2)^4}{4^2(4)} + \frac{(0.2)^6}{4^3(6)} - \frac{(0.2)^8}{4^4(8)} - 0$$

MATH 152
Mrs. Bonny Tighe

QUIZ 9

25 points
11.9

NAME Solutions

TA_____ Wed 11/16/2022

1. Find the power series representation for the following functions.

5pts

a) $f(x) = \dfrac{x^2}{1+4x^2}$ Find the interval of convergence for this series only.

$$\frac{x^2}{1-(-4x^2)} = \sum_{n=0}^{\infty} x^2(-4x^2)^n = \sum_{n=0}^{\infty} (-1)^n 4^n x^{2n+2}$$

$$|r| = |-4x^2| < 1 \qquad |4x^2| < 1 \qquad |x^2| < 1/4 \qquad |x| < 1/2$$

$$-1/2 < x < 1/2$$

$$\boxed{\begin{aligned} R &= 1/2 \\ I &= (-1/2, 1/2) \end{aligned}}$$

Don't need to test endpoints because it is a geometric series

3pts

b) $f(x) = \ln(x+3)$

$$f'(x) = \frac{1}{x+3} = \frac{1/3}{1+x/3} = \frac{1/3}{1-(-x/3)} = \sum_{n=0}^{\infty} \frac{1}{3}\left(-\frac{x}{3}\right)^n$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{3^{n+1}}$$

$$\int f'(x)\,dx = f(x) \qquad \int \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{3^{n+1}}\,dx =$$

$$C + \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{3^{n+1}(n+1)}$$

$$(C = \ln 3)$$

$u = 2x+1$
$du = 2dx$

3pts

c) $f(x) = \dfrac{1}{(2x+1)^2}$ $\quad \int f(x)dx \quad \int (2x+1)^{-2}dx$

$\frac{1}{-1}(2x+1)^{-1}\left(\frac{1}{2}\right) = \dfrac{-1/2}{1+2x} = \dfrac{-1/2}{1-(-2x)} = \sum_{n=0}^{\infty} (-1/2)(-2x)^n$

$\frac{d}{dx}\sum_{n=0}^{\infty} (-1)^{n+1} x^n 2^{n-1} = \boxed{\sum_{n=1}^{\infty} (-1)^{n+1} n x^{n-1} 2^{n-1}}$

Integrate, then differentiate

4pts

2. a) Approximate the integral using power series and 4 non-zero terms.

$\int_0^{0.2} \dfrac{2x^2}{4-2x^2}dx$ $\quad \int_0^{0.2} \dfrac{\frac{x^2}{2}}{1-\frac{x^2}{2}}dx = \int_0^{0.2} \sum_{n=0}^{\infty} \frac{x^2}{2}\left(\frac{x^2}{2}\right)^n dx$

$\int_0^{0.2} \sum_{n=0}^{\infty} \frac{x^{2n+2}}{2^{n+1}}dx = \sum_{n=0}^{\infty} \frac{x^{2n+3}}{2^{n+1}(2n+3)}\Big]_0^{0.2}$

$\frac{x^3}{2(3)} + \frac{x^5}{2^2(5)} + \frac{x^7}{2^3(7)} + \frac{x^9}{2^3(9)}\Big]_0^{0.2}$

$\left(\frac{(0.2)^3}{2(3)} + \frac{(0.2)^5}{2^2(5)} + \frac{(0.2)^7}{2^3(7)} + \frac{(0.2)^9}{2^3(9)}\right) - 0$

5 pts

3. Express the function as the sum of power series by first using partial fractions. Find the interval and radius of convergence. $f(x) = \dfrac{5+x}{3-10x+3x^2}$ $= \dfrac{5+x}{3x^2-10x+3}$

$(3x-1)(x-3)$

$\dfrac{A}{3x-1} + \dfrac{B}{x-3} = \dfrac{x+5}{(3x-1)(x-3)}$

$A(x-3) + B(3x-1) = x+5$

$x=3$ $\quad 3(8) = 8 \quad B=1$

$x = 1/3$ $\quad A(\frac{1}{3}-3) = \frac{1}{3}+5 = 16/3$

$A(-8/3) \quad A = \dfrac{16/3}{-8/3} = -2$

$\dfrac{1}{x-3} + \dfrac{-2}{3x-1}$

$\dfrac{-1}{3-x} + \dfrac{2}{1-3x}$

$\dfrac{-1/3}{1-x/3}$

$\sum_{n=0}^{\infty} -1/3\left(\frac{x}{3}\right)^n + \sum_{n=0}^{\infty} 2(3x)^n$

$\sum_{n=0}^{\infty} -\dfrac{x^n}{3^{n+1}} + 2(3)^n x^n$

$\boxed{\sum_{n=0}^{\infty} x^n\left[2(3^n) - \dfrac{1}{3^{n+1}}\right]}$

$|r| = |x/3|$ for one

$|r| = |3x|$ for the other

$\left|\frac{x}{3}\right| < 1 \quad |x| < 3 \quad I = -3, 3$

$|3x| < 1 \quad |x| < 1/3 \quad I = -1/3, 1/3$

$\boxed{\begin{array}{l} I \text{ for both is } (-1/3, 1/3) \\ R = 1/3 \end{array}}$

4. Find the derivative of the given power series

$$f(x) = \sum_{k=0}^{\infty} \frac{(-1)^k (1 \cdot 3 \cdot 5 \cdots (2k+1))\ x^{2k+1}}{3^{k+1}\ k!}$$

1pt $f'(x) = \sum_{k=0}^{\infty} \frac{(-1)^k (1 \cdot 3 \cdot 5 \cdots (2k+1))(2k+1)\ x^{2k}}{3^{k+1}\ k!}$

$$f(x) = \sum_{k=0}^{\infty} \frac{(-1)^k (1 \cdot 3 \cdot 5 \cdots (2k+1))\ x^{2k}}{7^k\ k!}$$

1pt $f'(x) = \sum_{k=1}^{\infty} \frac{(-1)^k (1 \cdot 3 \cdot 5 \cdots (2k+1))(2k)\ x^{2k-1}}{7^k\ k!}$

5. Find the integral of the power series $\int \sum_{n=0}^{\infty} \frac{(-1)^{n+1} (x+2)^{n+3}}{2^n} dx$

1pt $C + \sum_{n=0}^{\infty} \frac{(-1)^{n+1} (x+2)^{n+4}}{2^n (n+4)}$

6. Rewrite the series $f(x) = \sum_{n=1}^{\infty} \frac{(-1)^n\ x^{2n+1}}{3^{n+1}\ n!}$

2pts here $\sum_{n=0}^{\infty} \frac{(-1)^{n+1} x^{2n+2}}{3^{n+2}\ (n+1)!}$